



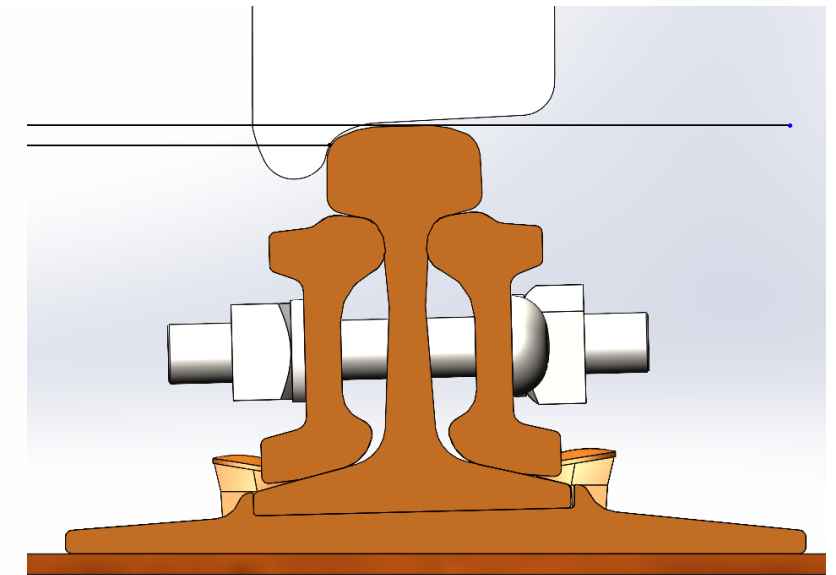
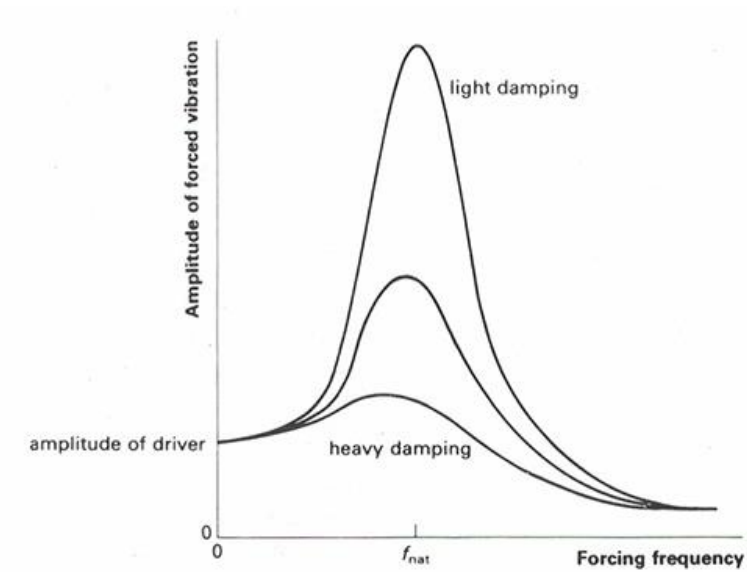
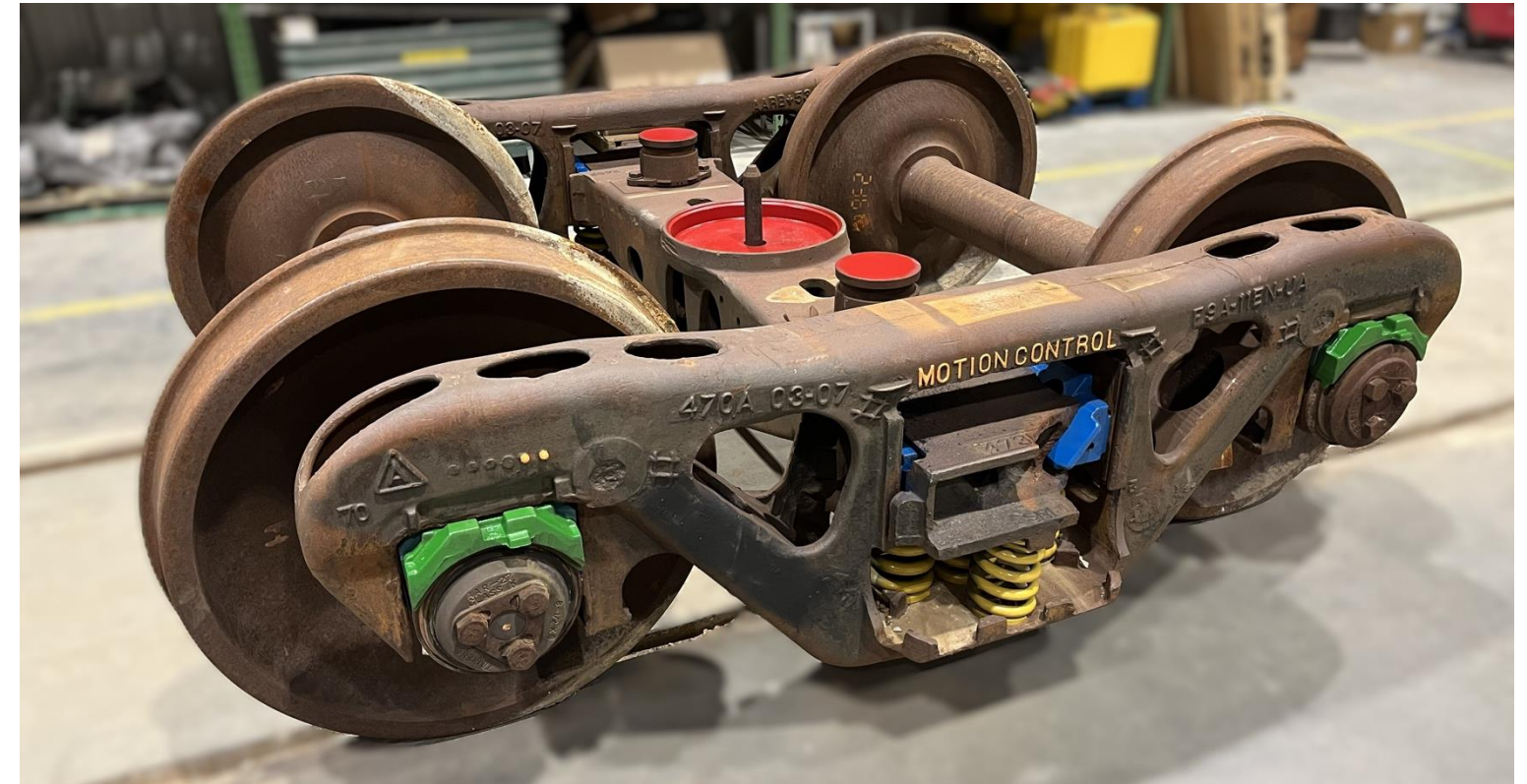
Vehicle Types, Suspensions and Components

DALLAS, TX



Overview

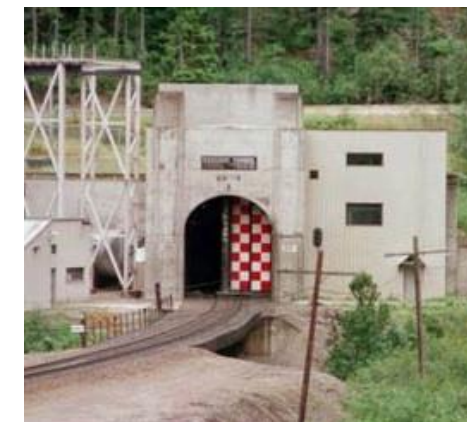
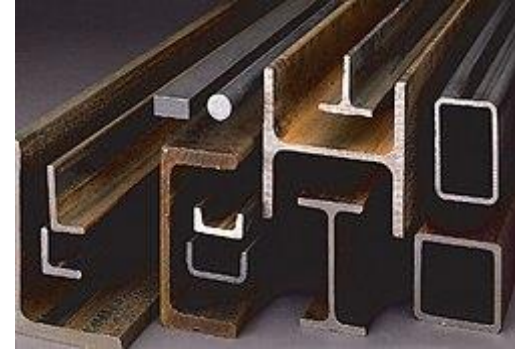
- Rail car types, sizes, weights, shapes
- Components and arrangements
- Three-piece freight truck
- Suspension and steering
- Resonance
- Tangent and curving performance





Rail Car Design

- **Designed by commodity**
 - Liquids/Gases, particles, raw material, finished goods
- **Handling at origin and destination**
 - Cranes, forklifts, chutes, pipes, magnets, grapples
- **Clearances**
 - Tunnels, wayside structures, in-track equipment
- **Allowable weights**
 - Soil condition, bridges, rail sections, crossties, ballast





Rail Car Designs – Car Types

Flat



Standard



Bulkhead



Centerbeam



Autorack

Box



Standard



Refrigerated / Insulated



Auto Parts



Rail Car Designs – Car Types

Tank



Non-Hazardous



Hazardous, Flammable



Hazardous, Non-Flammable



Refrigerated Gases

Hopper



Cement



Grain



Pressure Transfer

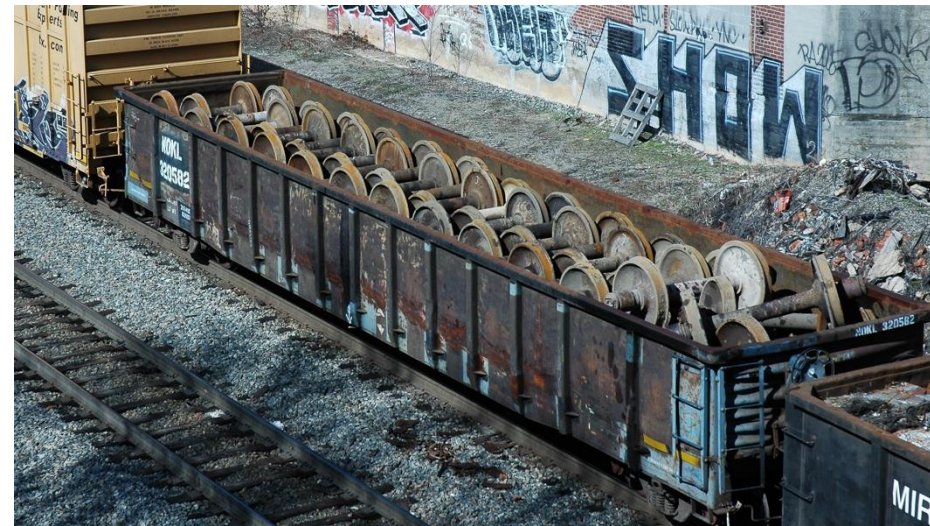


Open-Top Hopper



Rail Car Designs – Car Types

Gondola



Mill Gondola



Coal Gondola



Coil Gondola

Intermodal



Spine Car



Well Car



Road Railer



Rail Car Designs – Specialty Car Types



Schnabel / Kasgro *



Aircraft



Wind Turbine Blades



“Scorpion” Car for MoW *



Rail



Ties



Side Dump Gondola *



Nuclear Fuel

* Frank Orona Photos



Rail Car Designs – Clearances

• Wayside

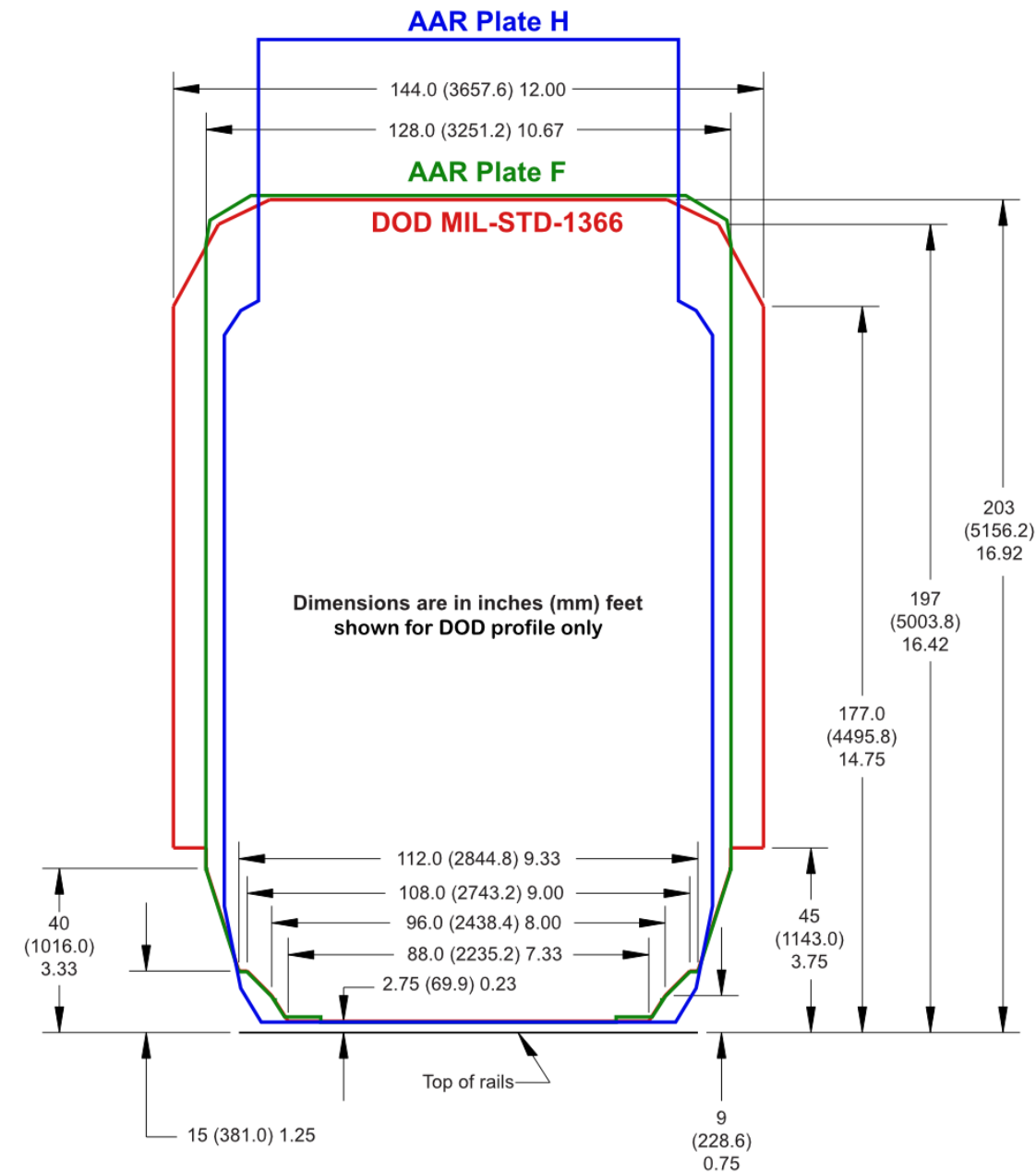
- Tunnels
- Signals
- Platforms

• Overhead

- Tunnels
- Bridges
- Catenary

• AAR “Plates”

- Width : 10' – 10'8"
- Height: 16' – 20'3"
- Overhang in curves





Rail Car Designs – Weights

- **GRL** = Gross Rail Load
- **LT WT** = Light Weight
- **LD LMT** = Load Limit
- **GRL determined by infrastructure**
 - 1960's - 263k, "100 ton", Big John case
 - 1990's - 286k, "110 ton"
 - 2000's - 315k, "125 ton" (select routes)
 - Isolated mining ops: 352k
 - **Net-to-Tare Ratio** : Loads 5-6x heavier than empties **





Catch up: Rail Car Design

Commodity

- Bulk / Finished
- Liquid, Solid, Gas



Handling

- Cranes, forklifts
- Chutes, pipes
- Scoops, magnets



Clearance

- Wayside
- Overhead
- Curving / Overhang



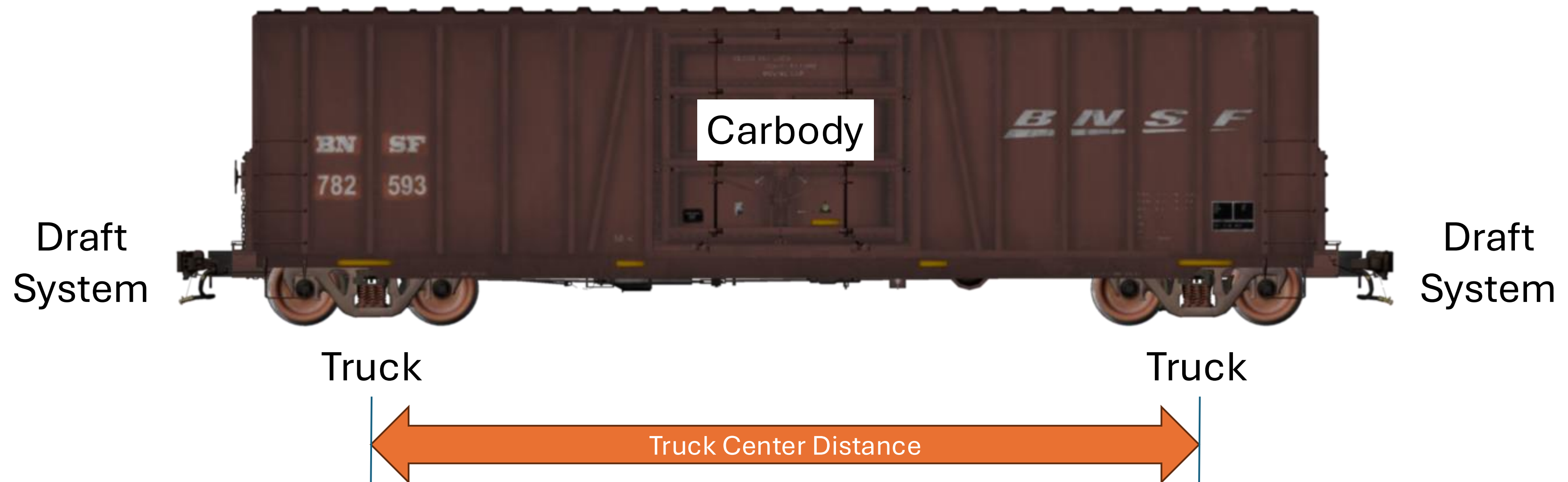
Infrastructure

- 286K GRL
- Rail, ties
- Soil, ballast, bridges



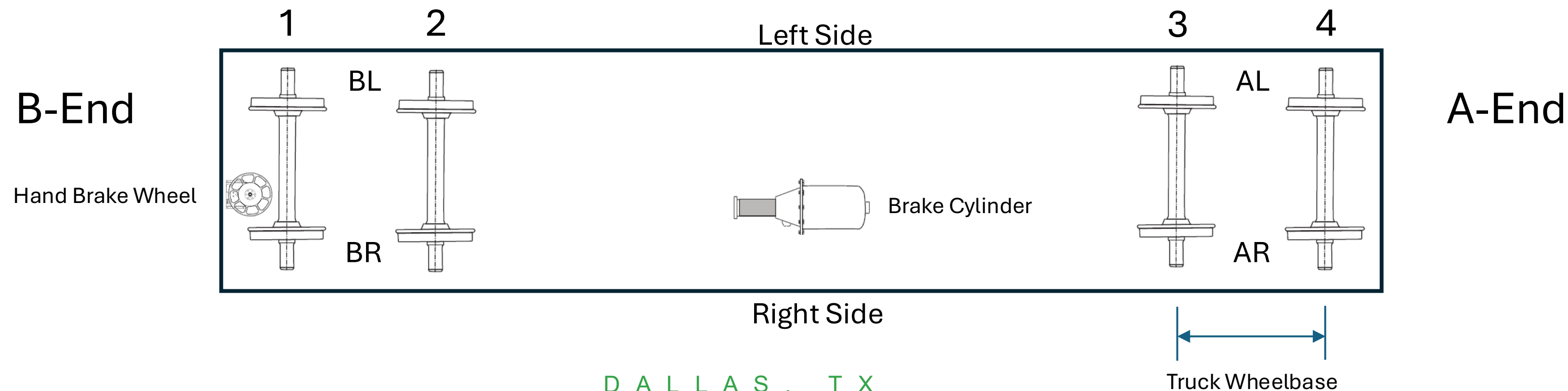


Arrangement - General





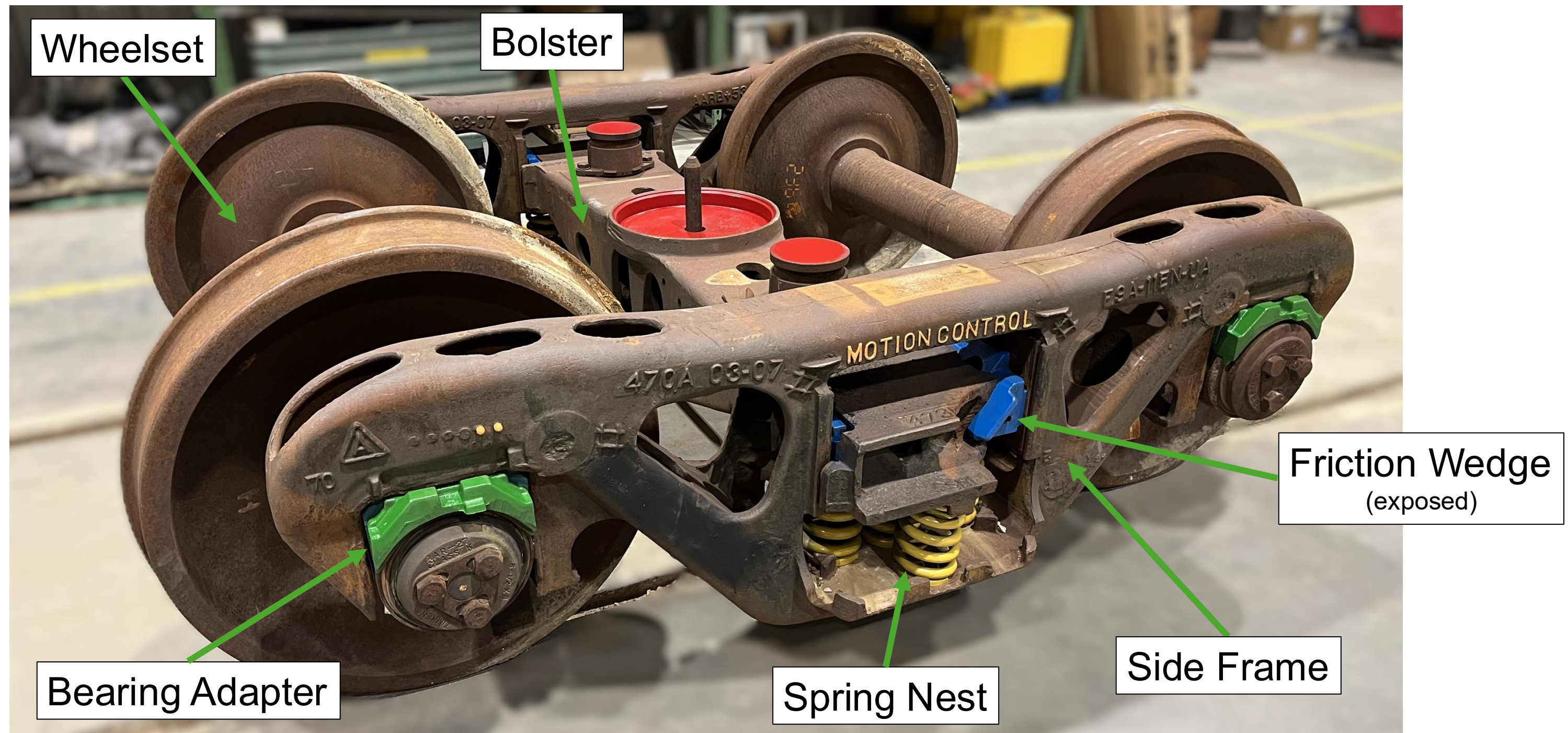
Arrangement - General



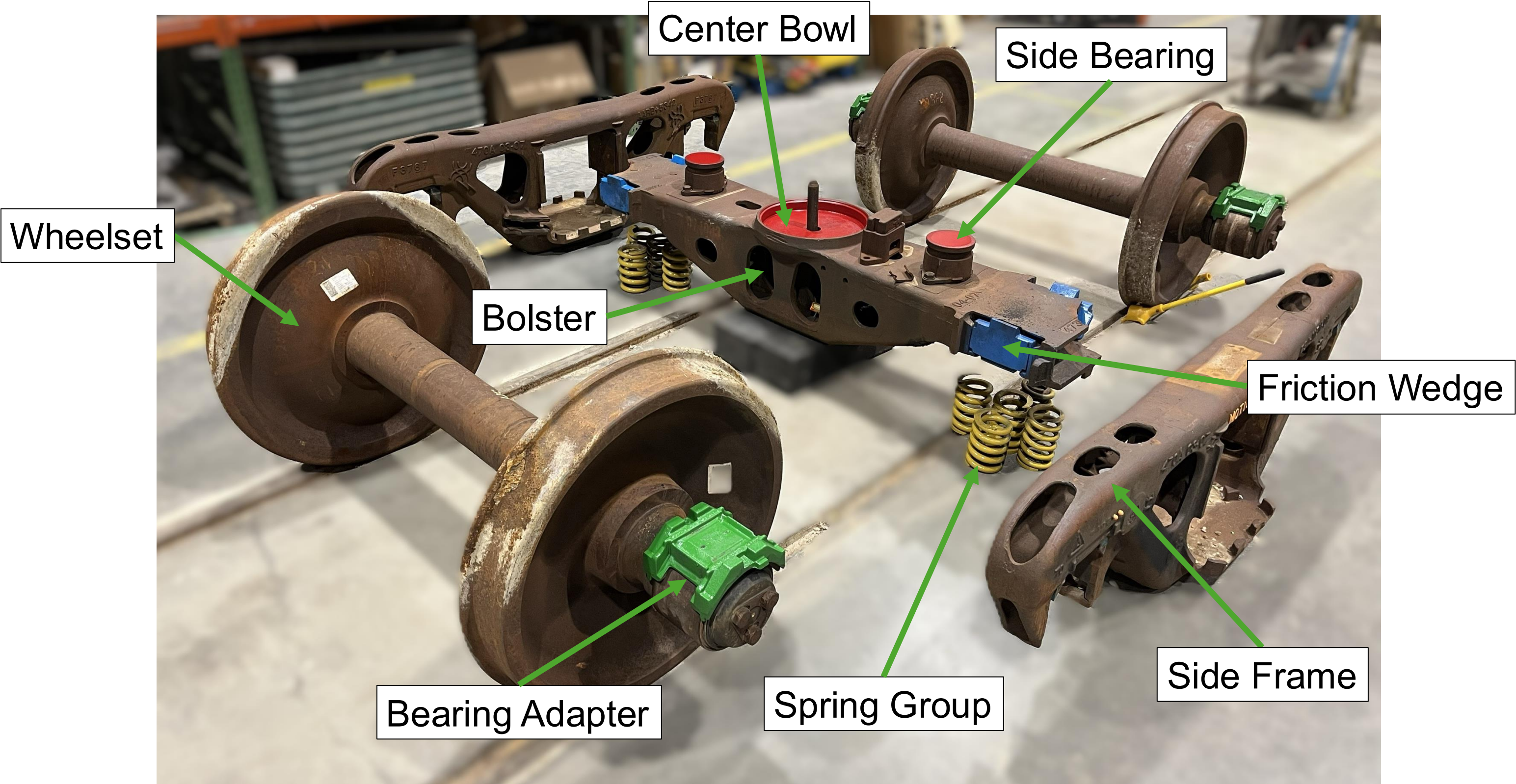




Arrangement - Truck



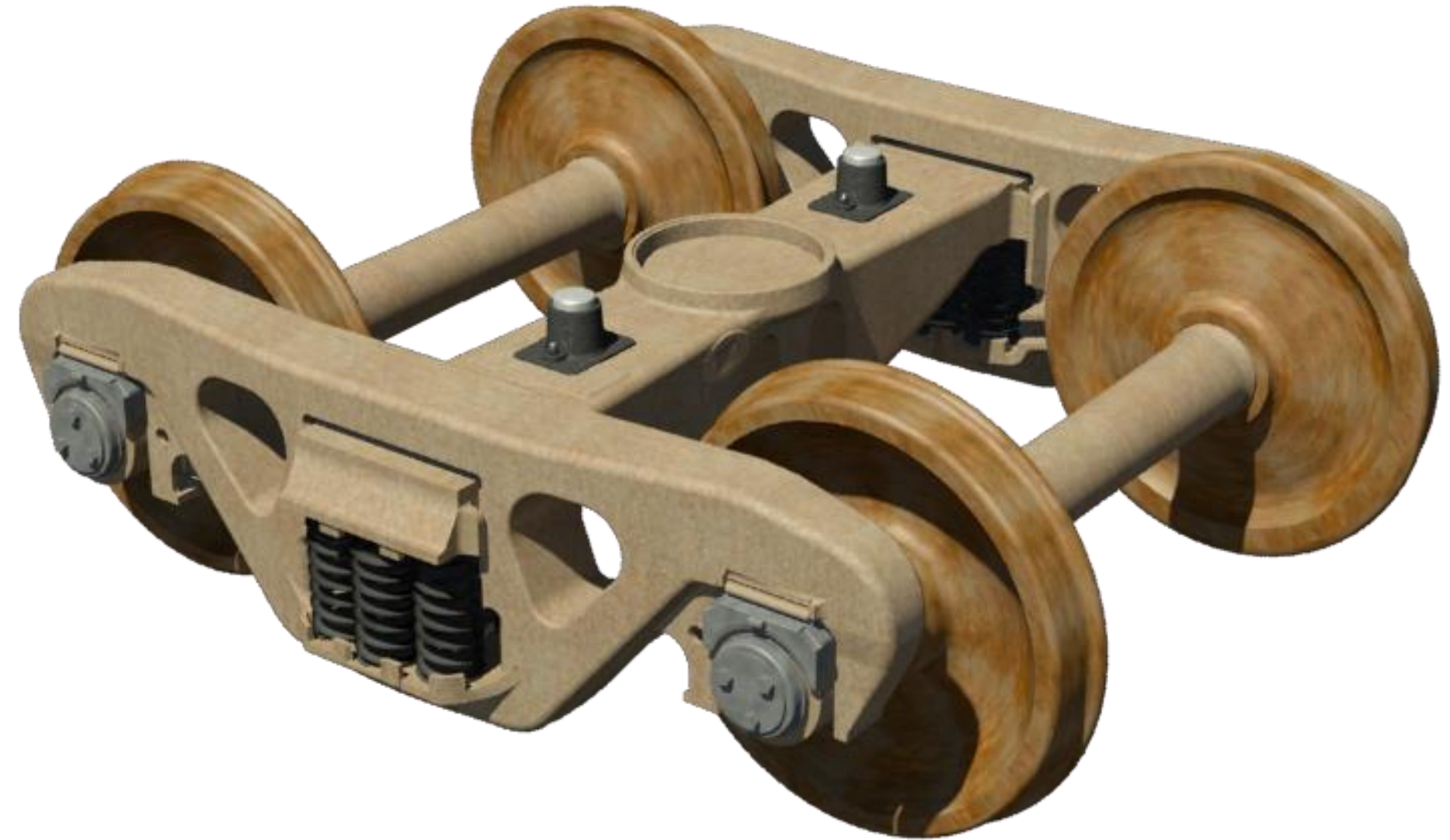
Arrangement - Truck





Three-piece truck

- Not high tech, but good tech
- Simple design, complex details
- Three functions
 - Keep wheels on the rail
 - Smooth and stable ride
 - Maintain intended direction





Three-piece truck

- Not high tech, but good tech
- Simple design, complex details
- Three functions
 - Stay upright and on the rail
 - Smooth and stable ride
 - Maintain intended direction



New, low miles, unworn



Used, high miles, worn in



Broken, worn out

Three-piece truck

Suspension and Steering

- **Vertical**, but affects lateral
 - Vertical load on wheels
 - Absorbs vertical shocks
 - Helps keep truck square

- **Lateral**, but affects vertical
 - Lateral forces on wheels
 - Absorbs lateral shocks
 - Keeps wheels aligned to the track





Catch up: Car Arrangement

Carbody

- Shaped by commodity
- Balance: GRL, length, clearance



Wheelset

- Only thing contacting the rail
- Simple shape
- Complex contact



Trucks

- Wheels on rails
- Smooth ride
- Run straight



Suspension and Steering

- Like your own vehicle
- Vertical
- Lateral
- Lat/Vert interaction

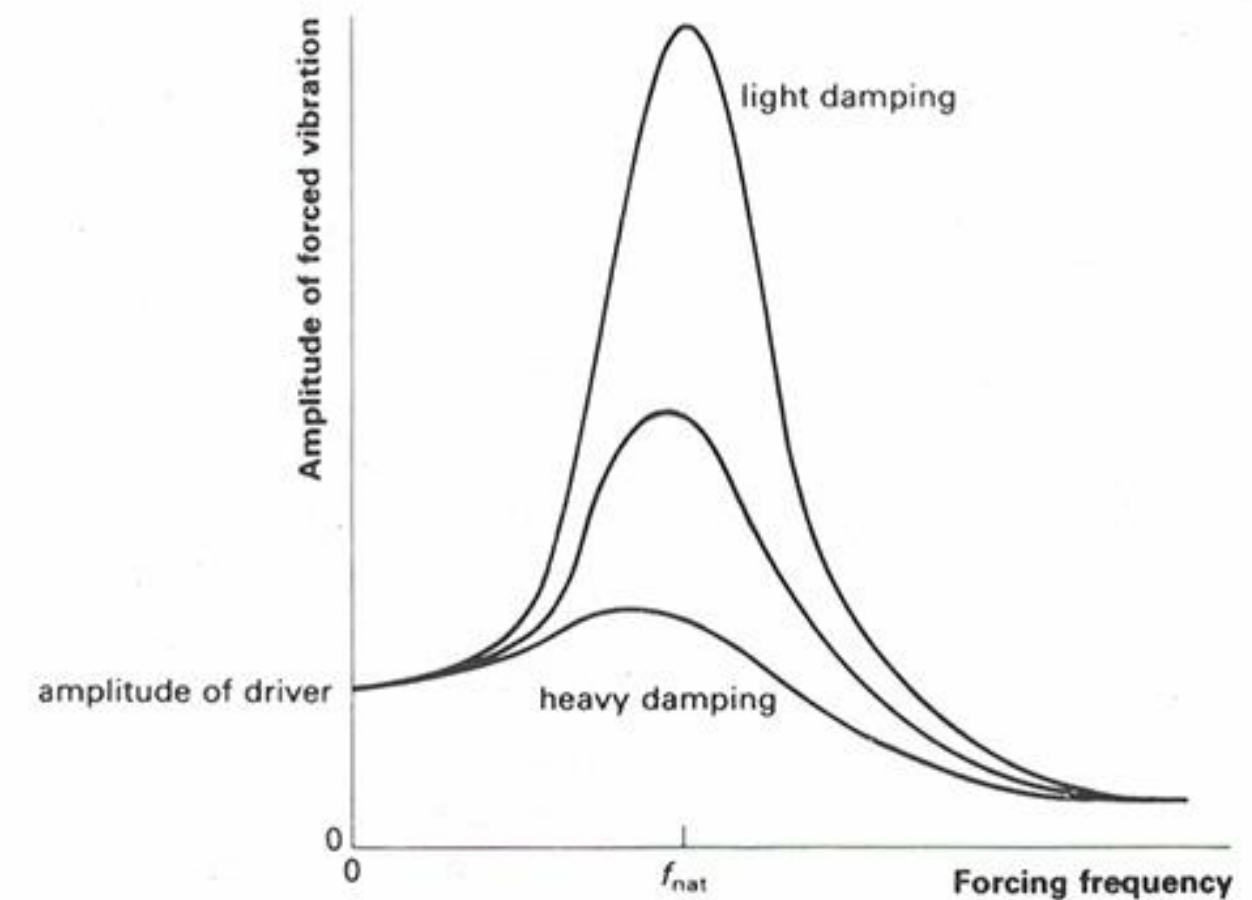
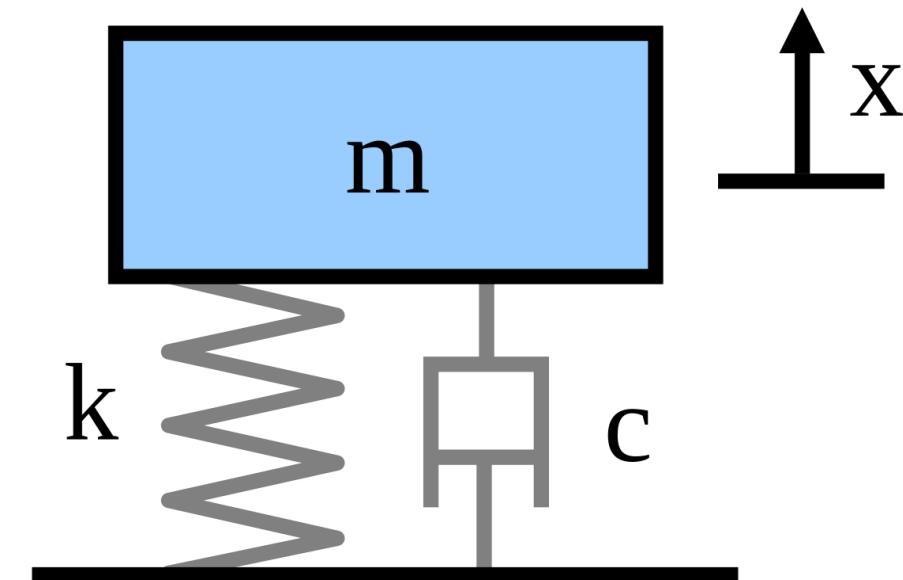




Suspension

- “Suspensions have springs”
 - Spring-mass-damper system
 - Keep the wheels on the rail
 - Maintain a smooth and stable ride
- Resonance
 - “Natural frequency”
 - Small inputs = Large outputs at specific freq’s.

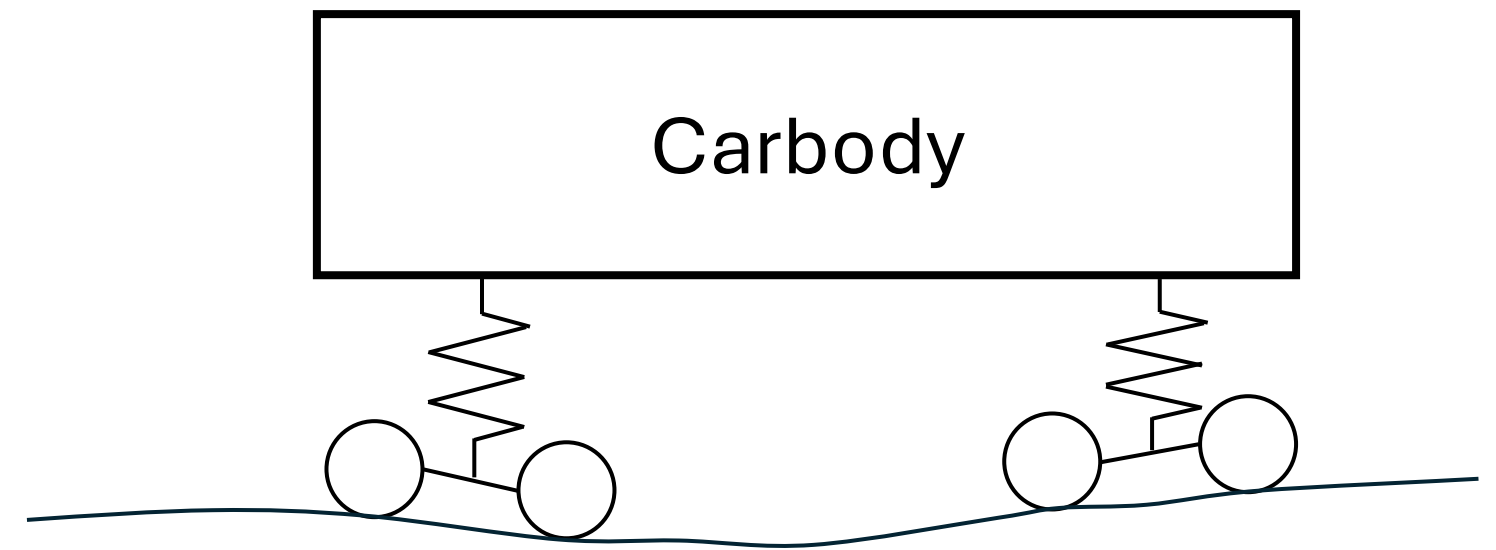
$$f_n = \sqrt{\frac{k}{m}}$$





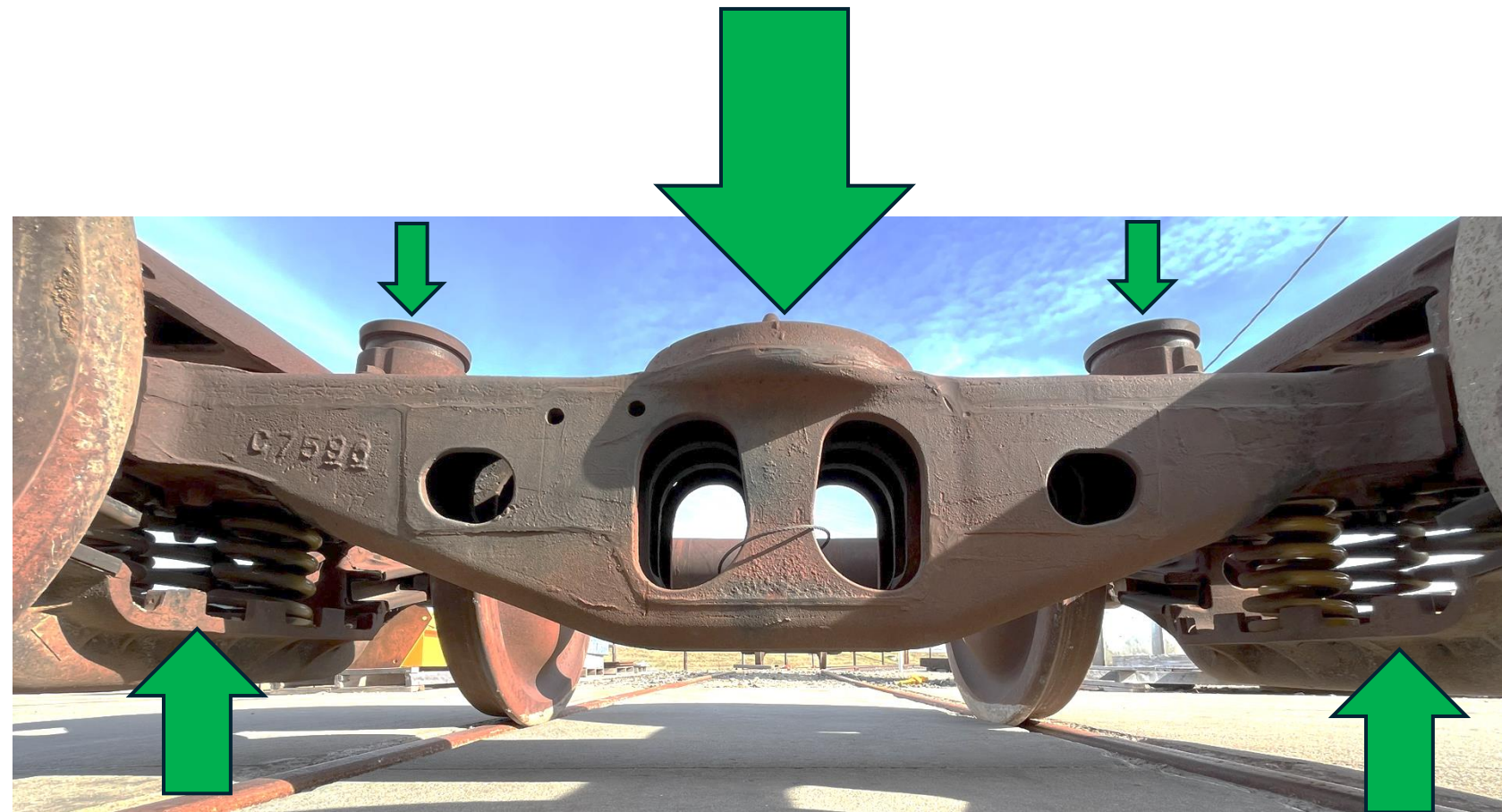
Suspension

- **Carbody** has most of the weight / mass
- **Springs** push wheels down to ground
- **Dampers** absorb energy from springs

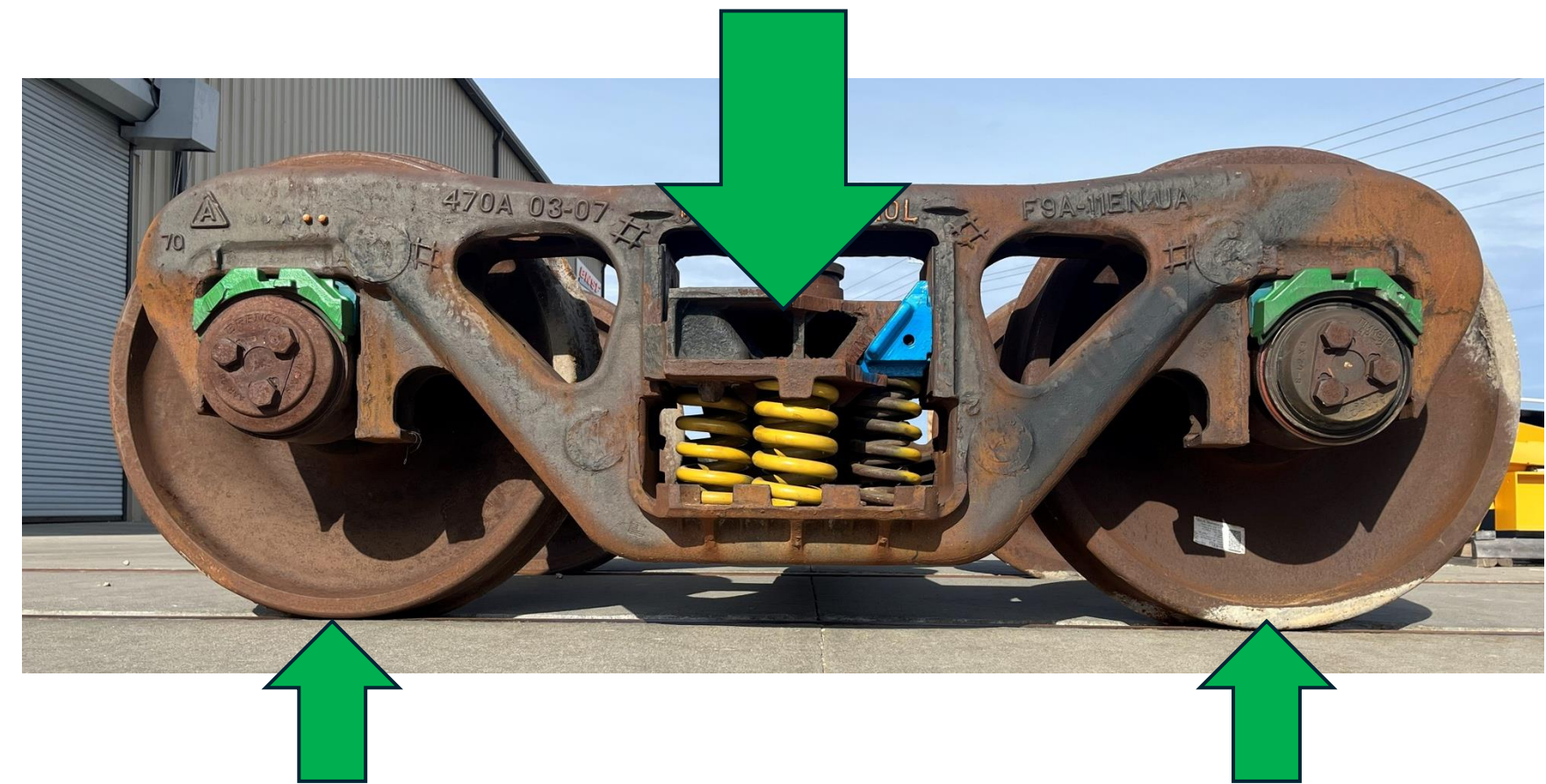




Suspension - Springs



Lateral Equalization

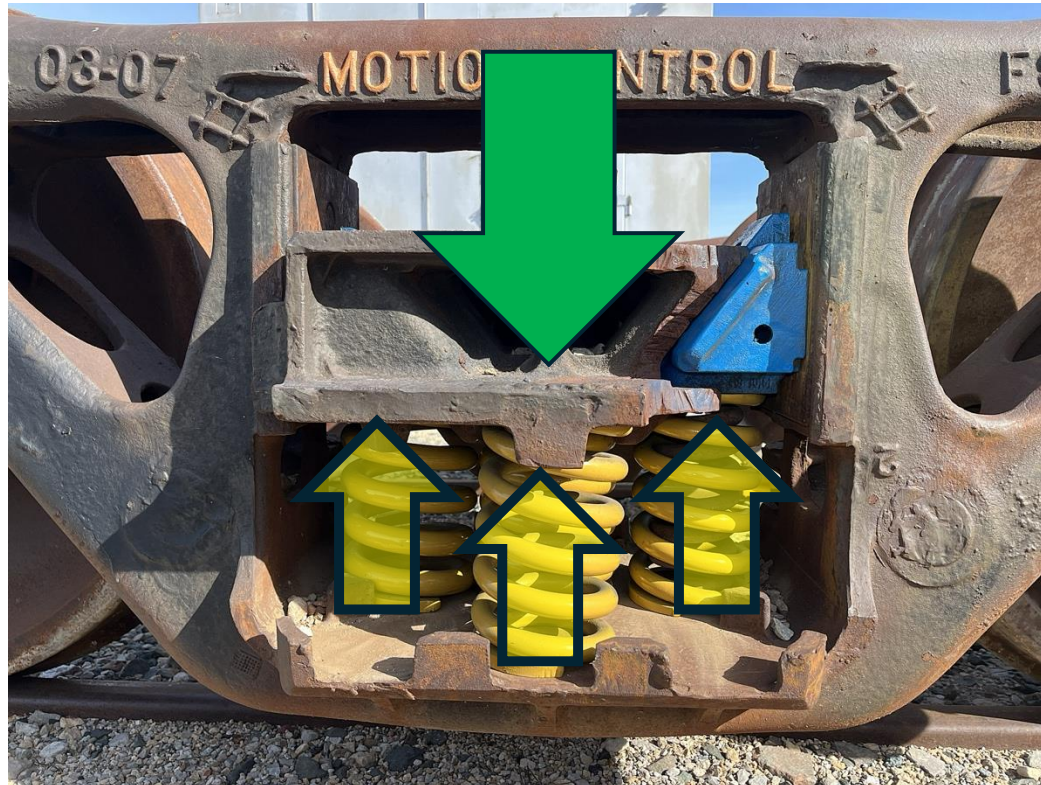


Longitudinal Equalization

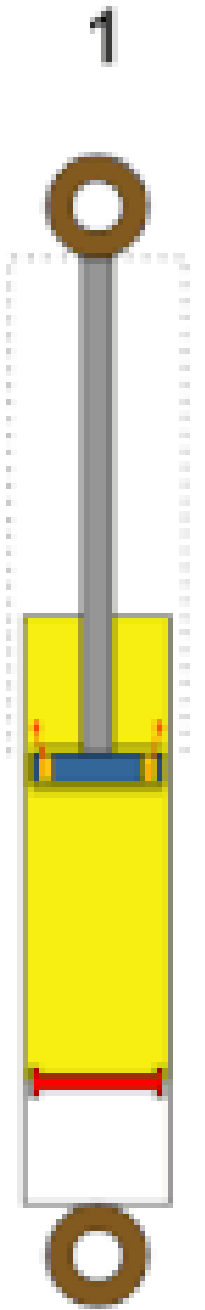
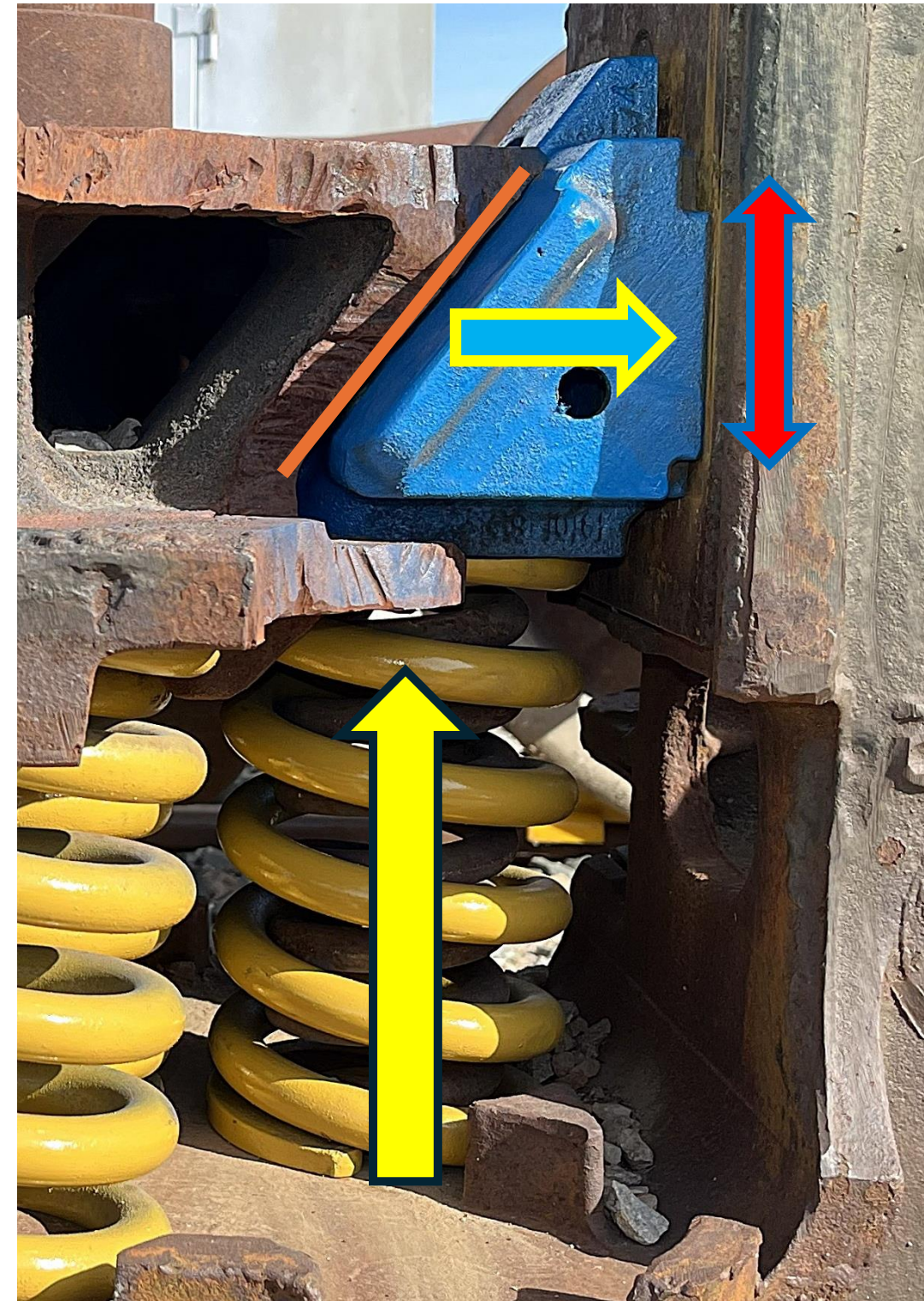


Suspension - Damping

- **Dampers = Friction Wedges**
- Remove unwanted energy
- Spring up - Wedge out
- Friction on column wear plate



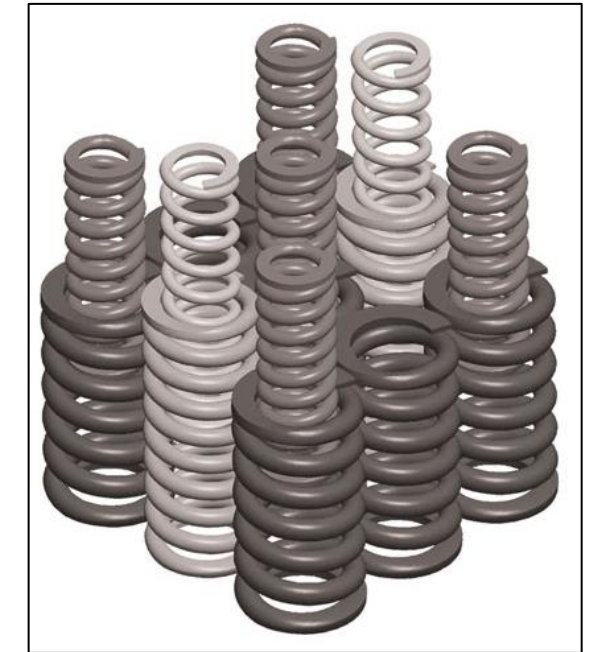
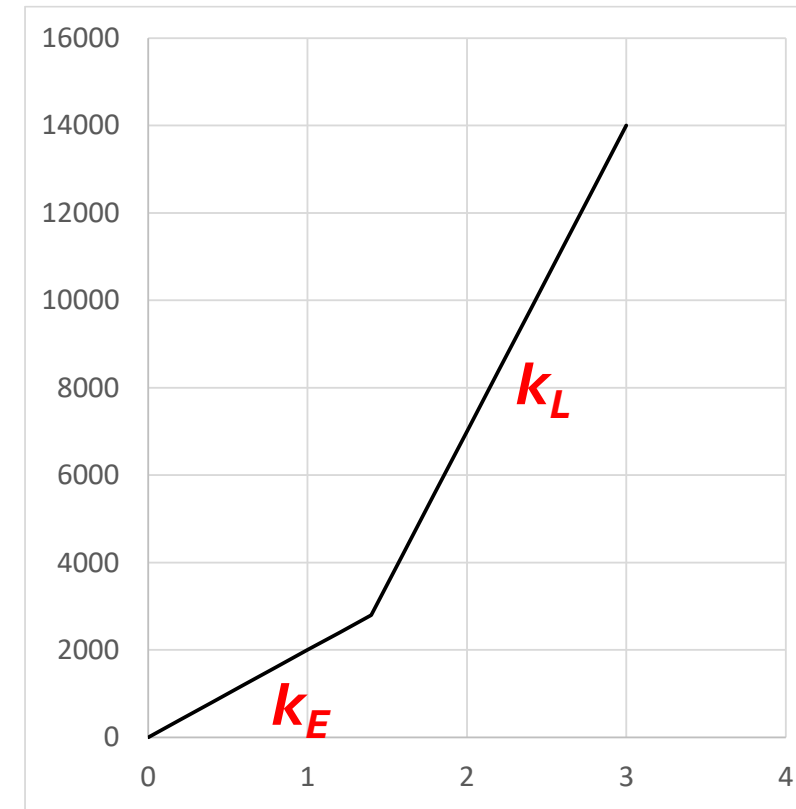
Cut away view of wedge



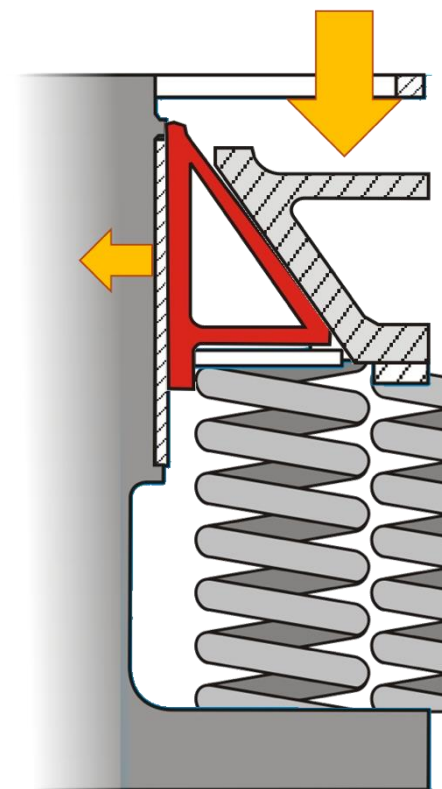


Suspension – Fine Tuning

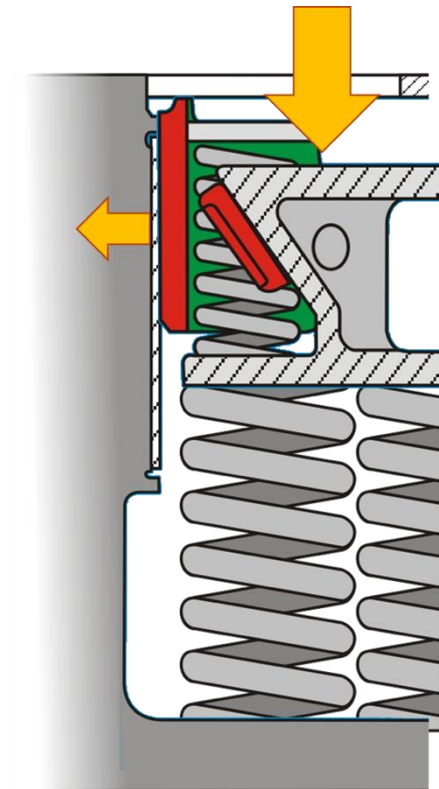
- **Design for load or empty?**
 - Both!
- **Springs**
 - Piecewise Linear Rate
 - Inner/Outer Coils
- **Dampers**
 - Variable damping
 - Constant damping



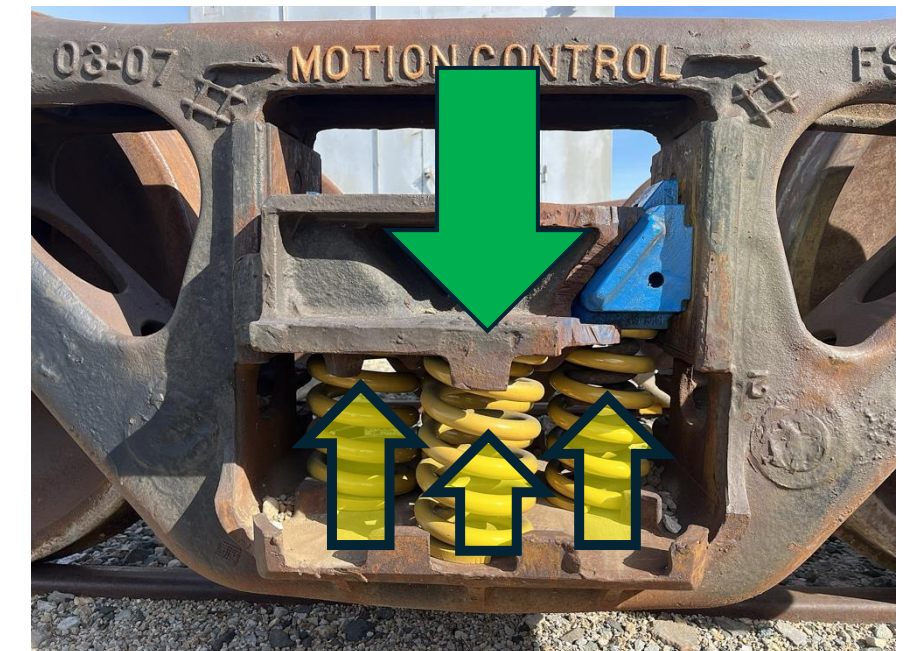
<https://www.railway-technology.com/contractors/bogies/amsted-rail/>



Variable Damped



Constant Damped





Suspension - Resonance

- Carbody roll
- Pitch, bounce, yaw

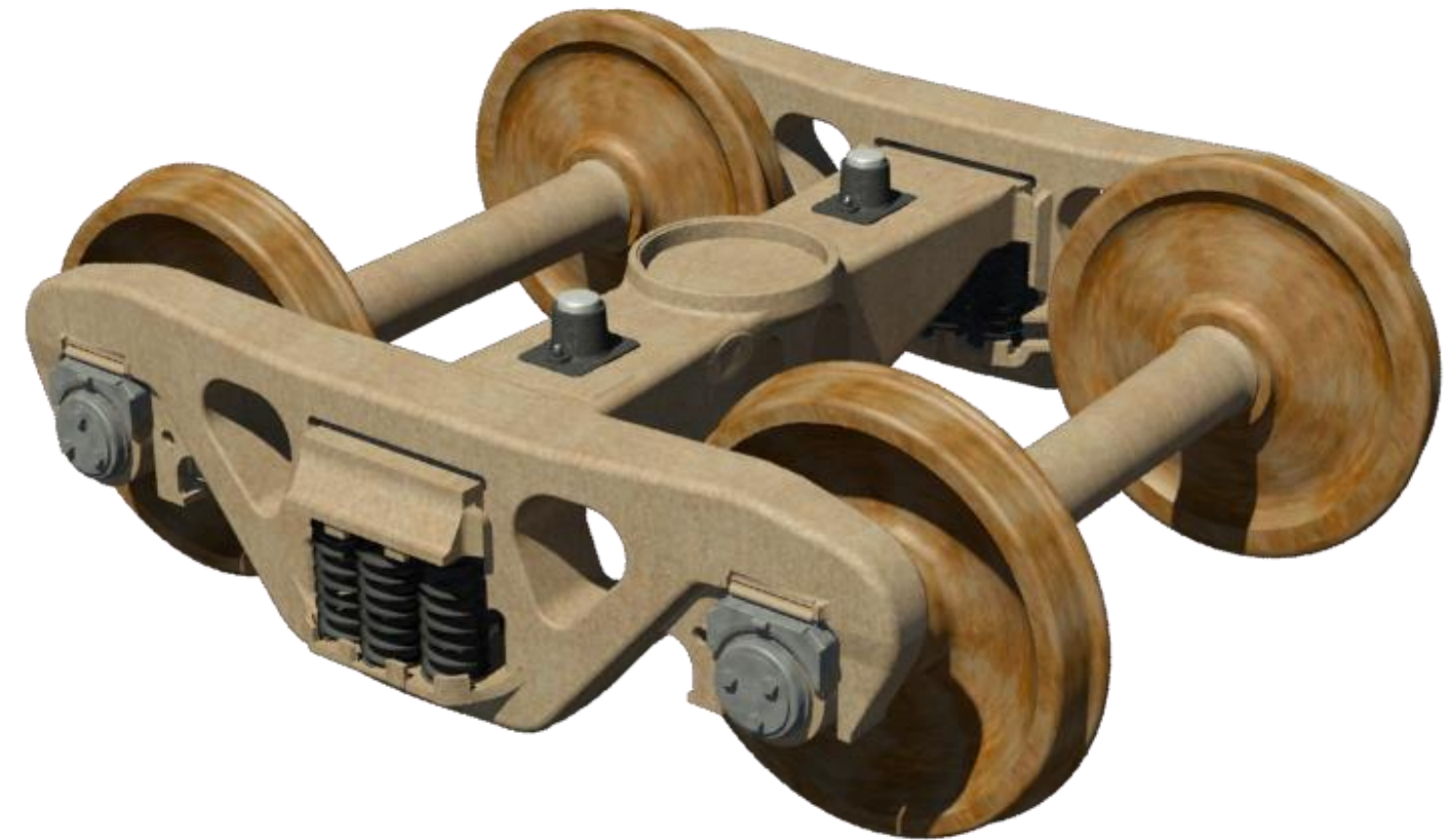


Video: Penn Central (1974)



Suspension to steering

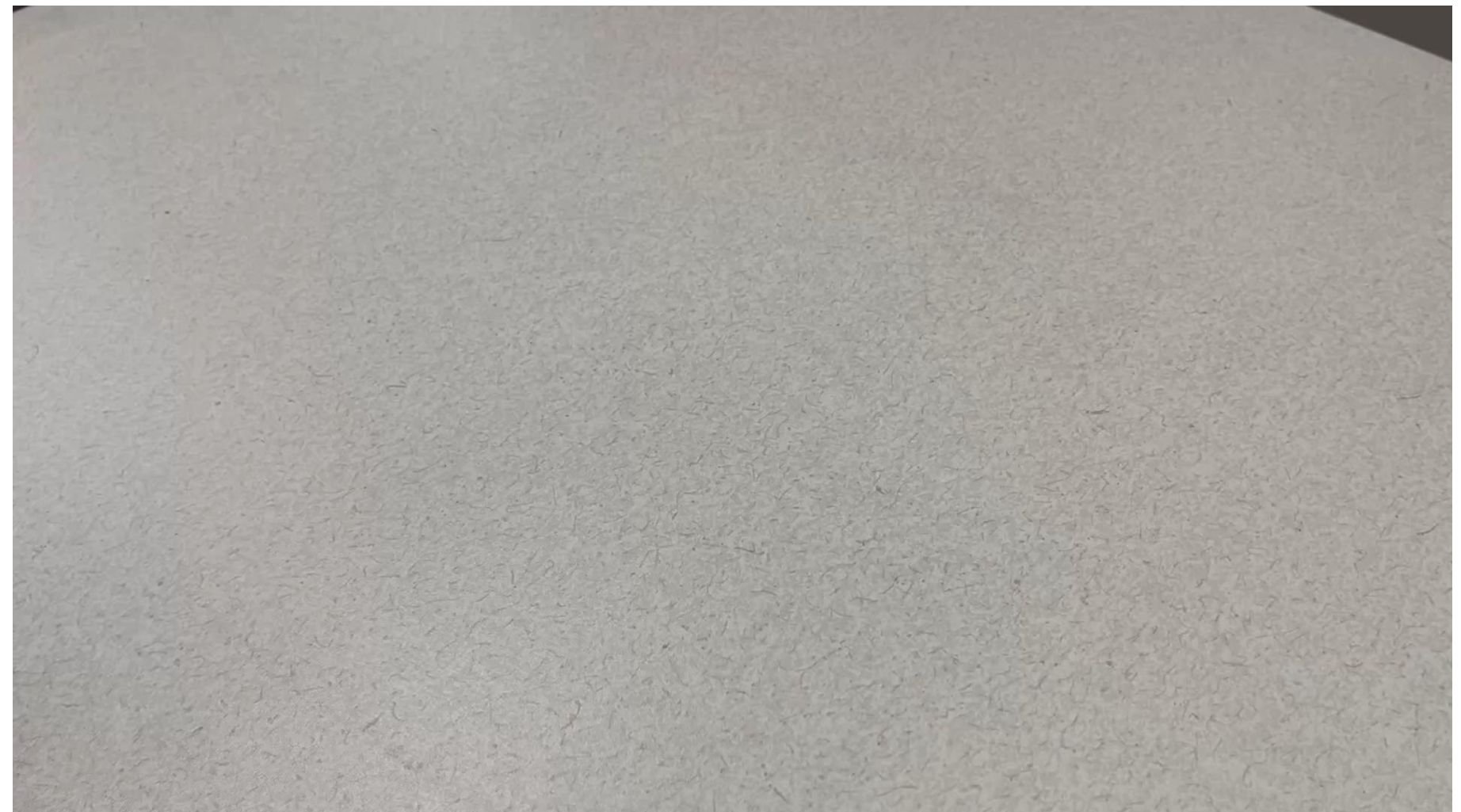
- **Catch up: Suspension**
 - Equalize load – stay upright and on the rail
 - Smooth ride – isolate shock and vibration
 - Springs and wedges
- **Lead in: Steering**
 - Taper of wheels
 - Rigidity of truck and wedge condition
 - Swivel stiffness





Steering – Wheel Taper

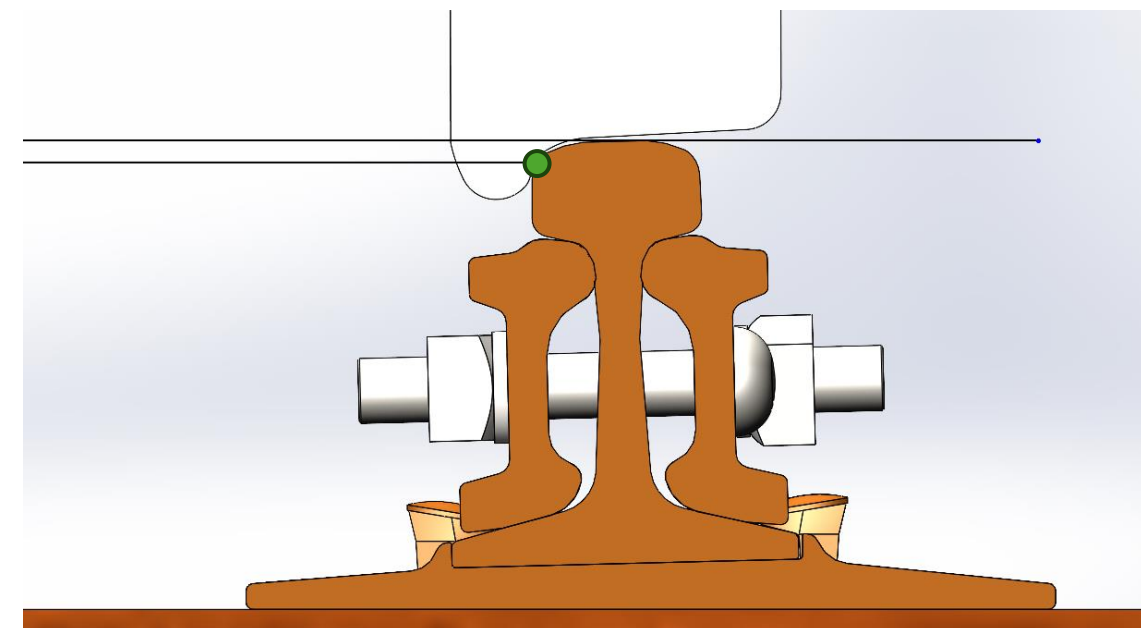
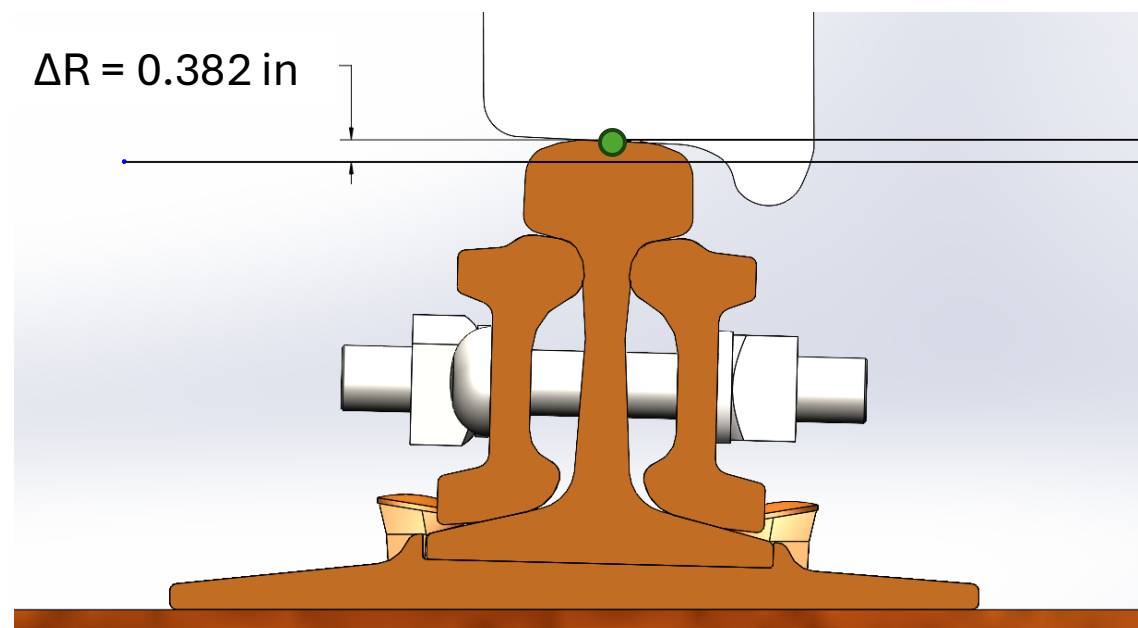
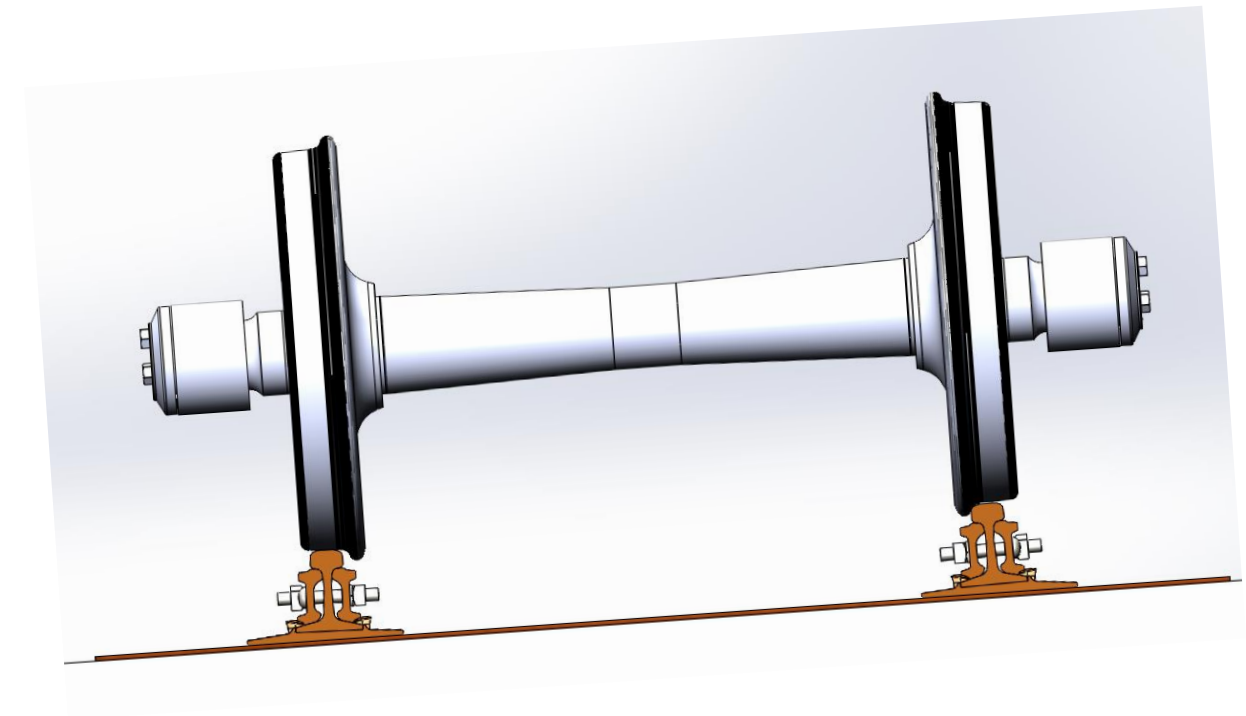
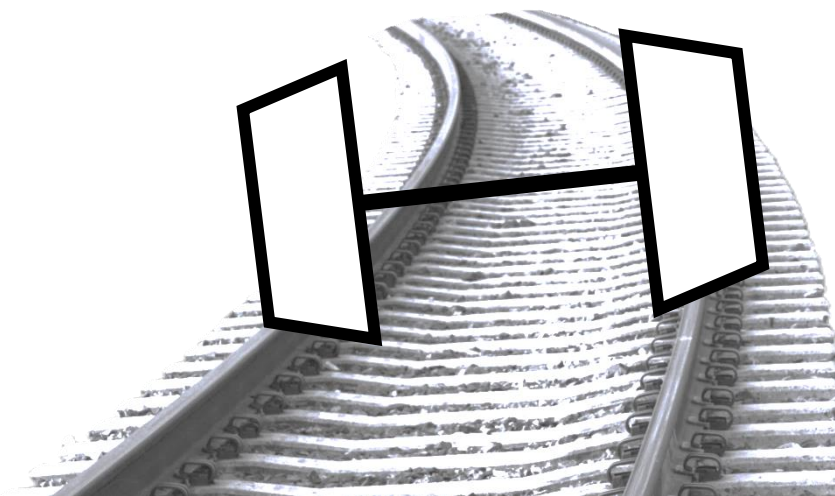
- **Coffee cup example**
 - Why does the cup roll in a circle to the left?





Steering – Wheel Taper

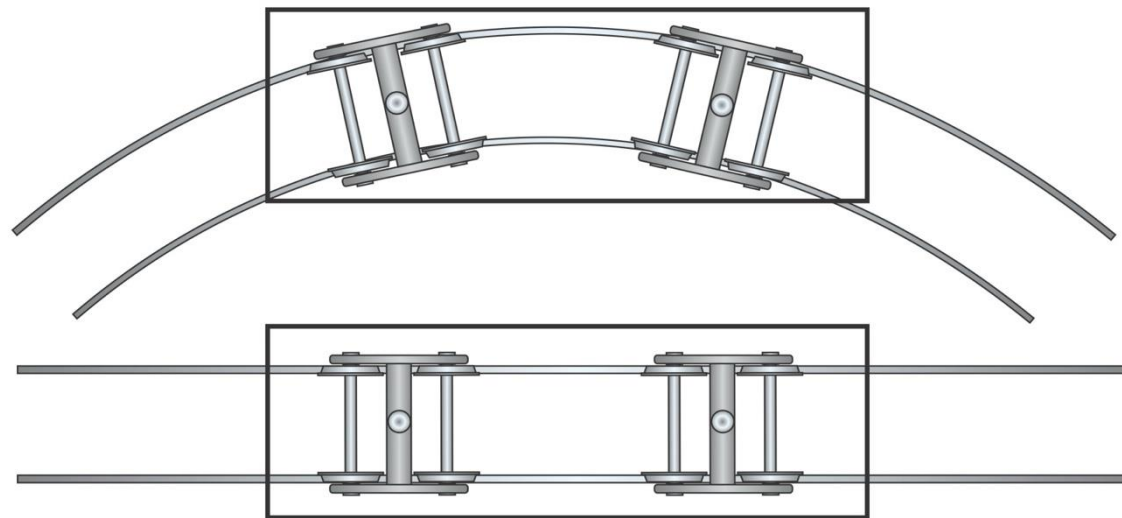
- Why does the cup roll in a circle?
 - Different diameters
- Wheelsets do the same



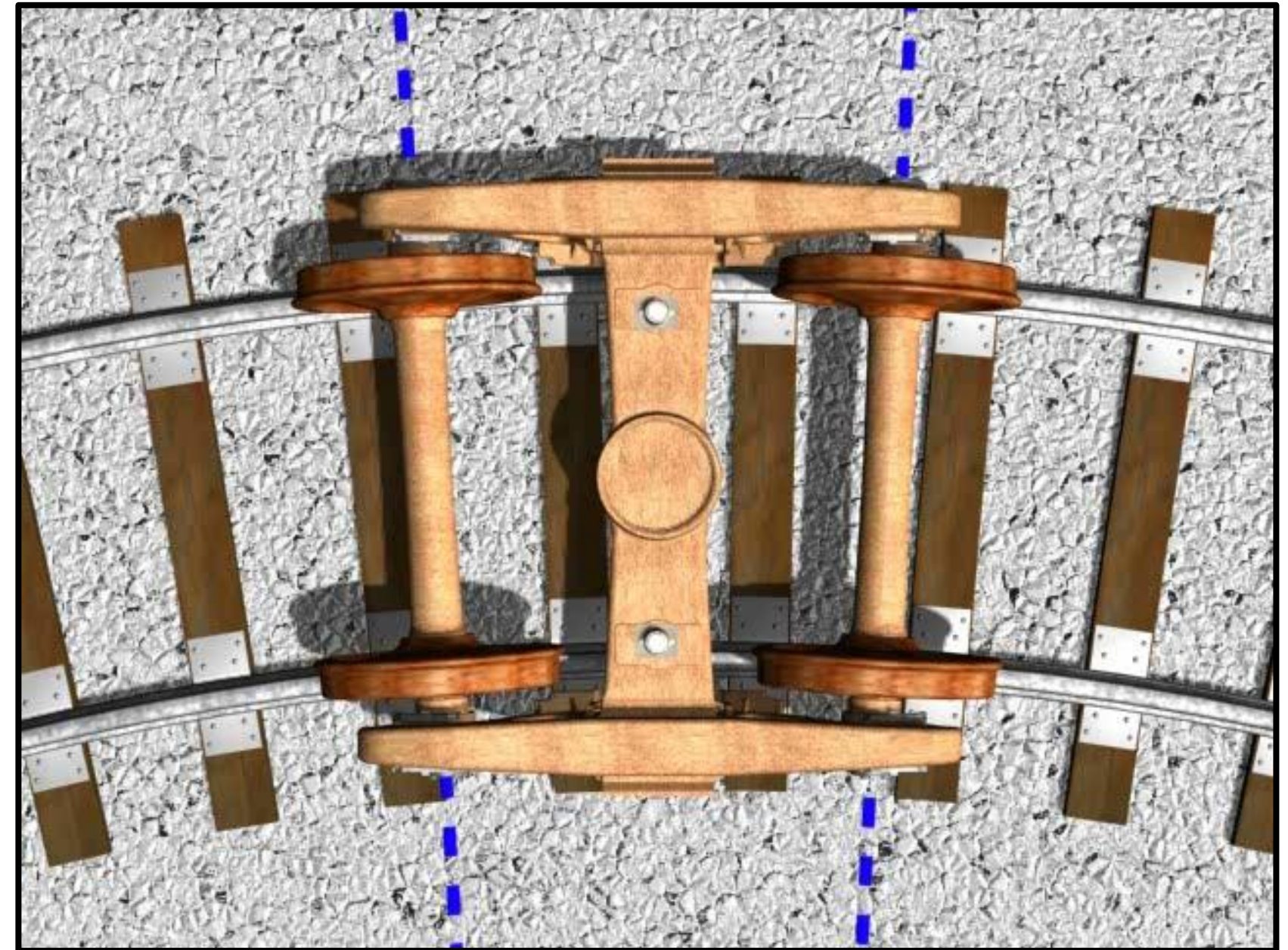


Steering – Truck Rigidity

- Wheelsets do the steering
- Truck warp
- Angle of attack
- Truck yaw / swivel



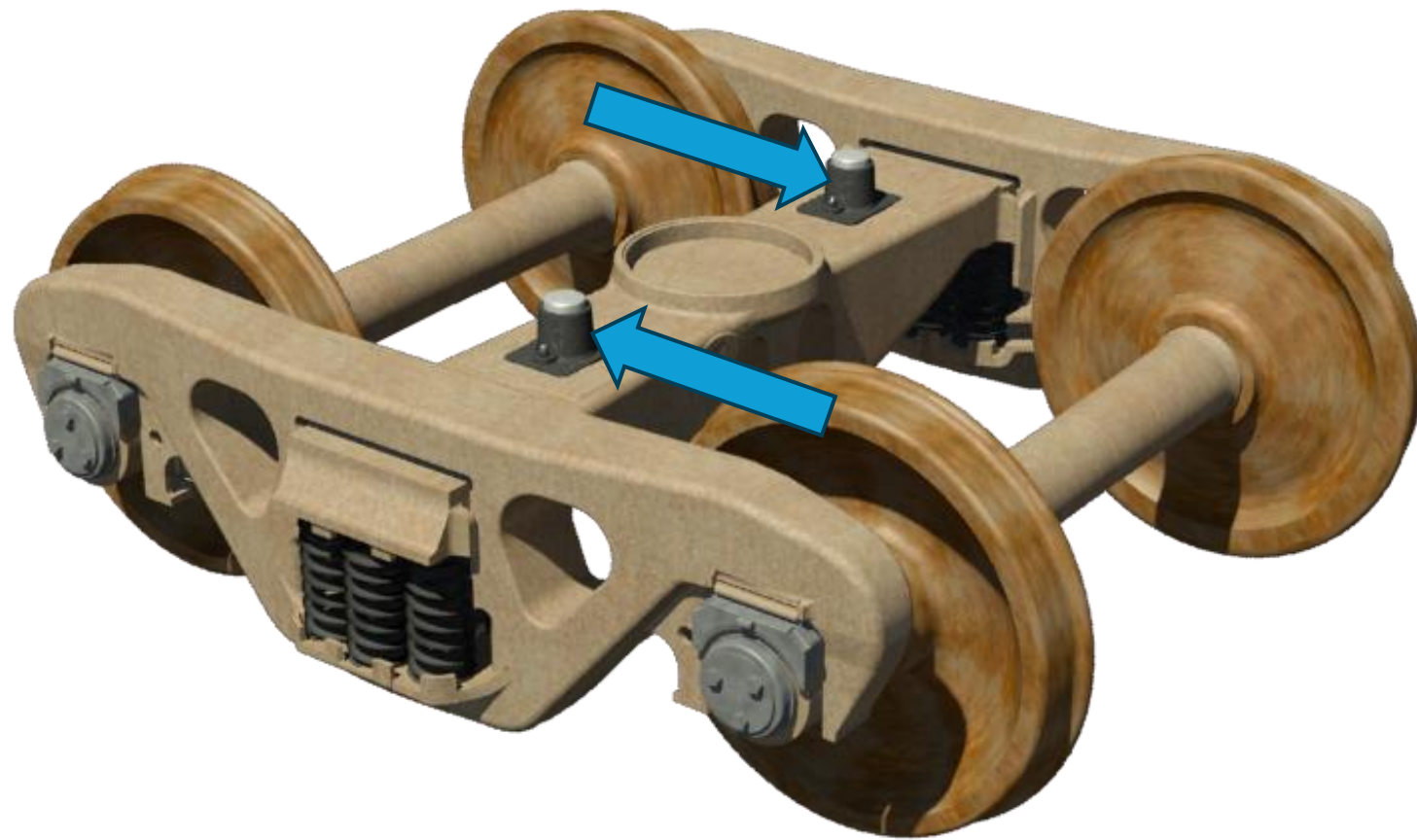
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Steering – Resonance

- **Hunting** = unwanted yaw resonance
- **High speed stability**
- **Damping** through side bearings





Catch up: Suspension and steering

Suspension

- Wheels on rail
- Smooth and stable ride
- Load upright



Steering

- Smooth and stable
- Keep pointed in the right direction



Interactions

- Lateral affects vertical
- Vertical affects lateral
- Resonance



Worn components

- Wedges
- Side bearings
- Center bowl swivel

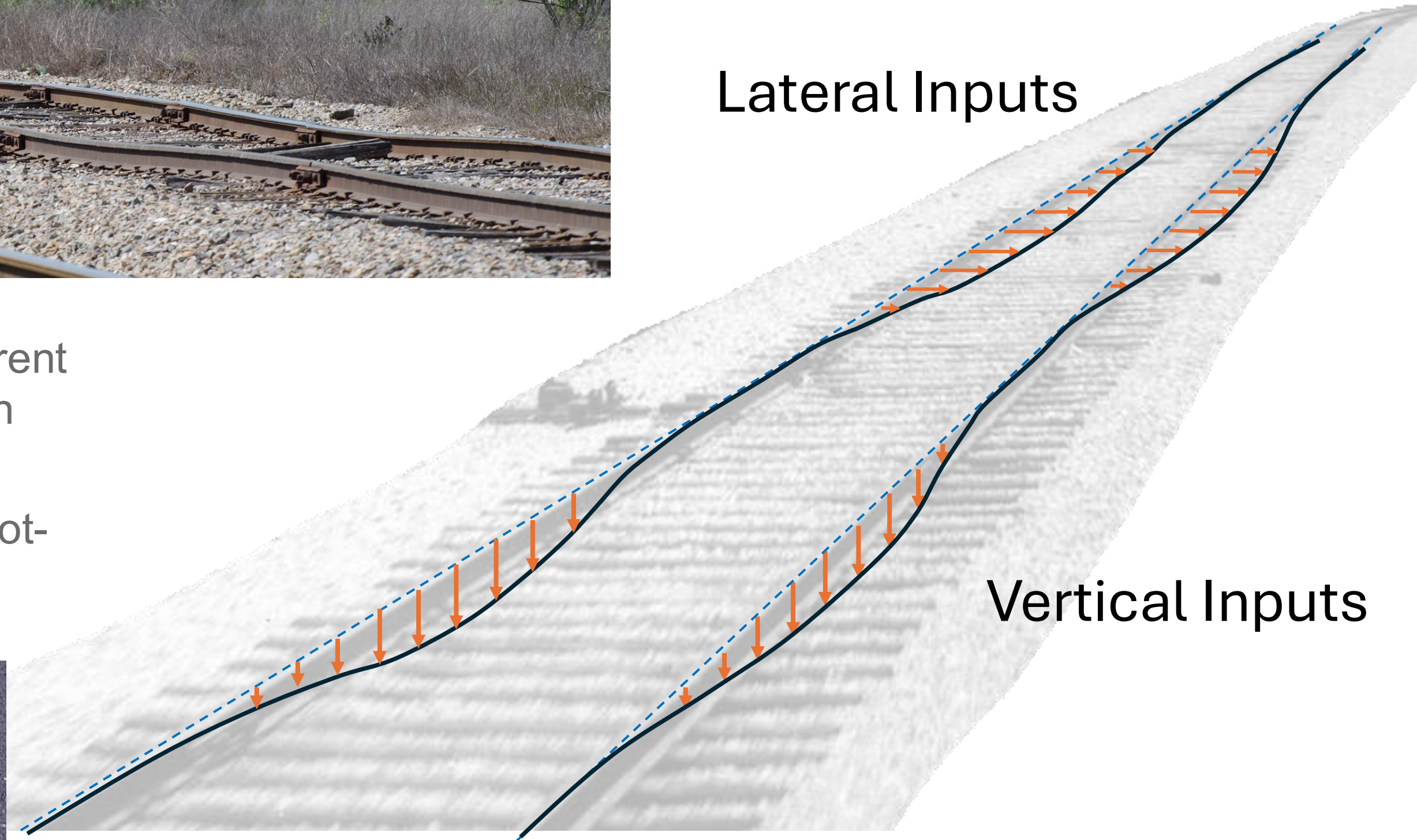




Tangent track



- **Expectation:** Navigate inherent track irregularities for a given class of track.
- **Example:** Rumble strip vs pot-hole

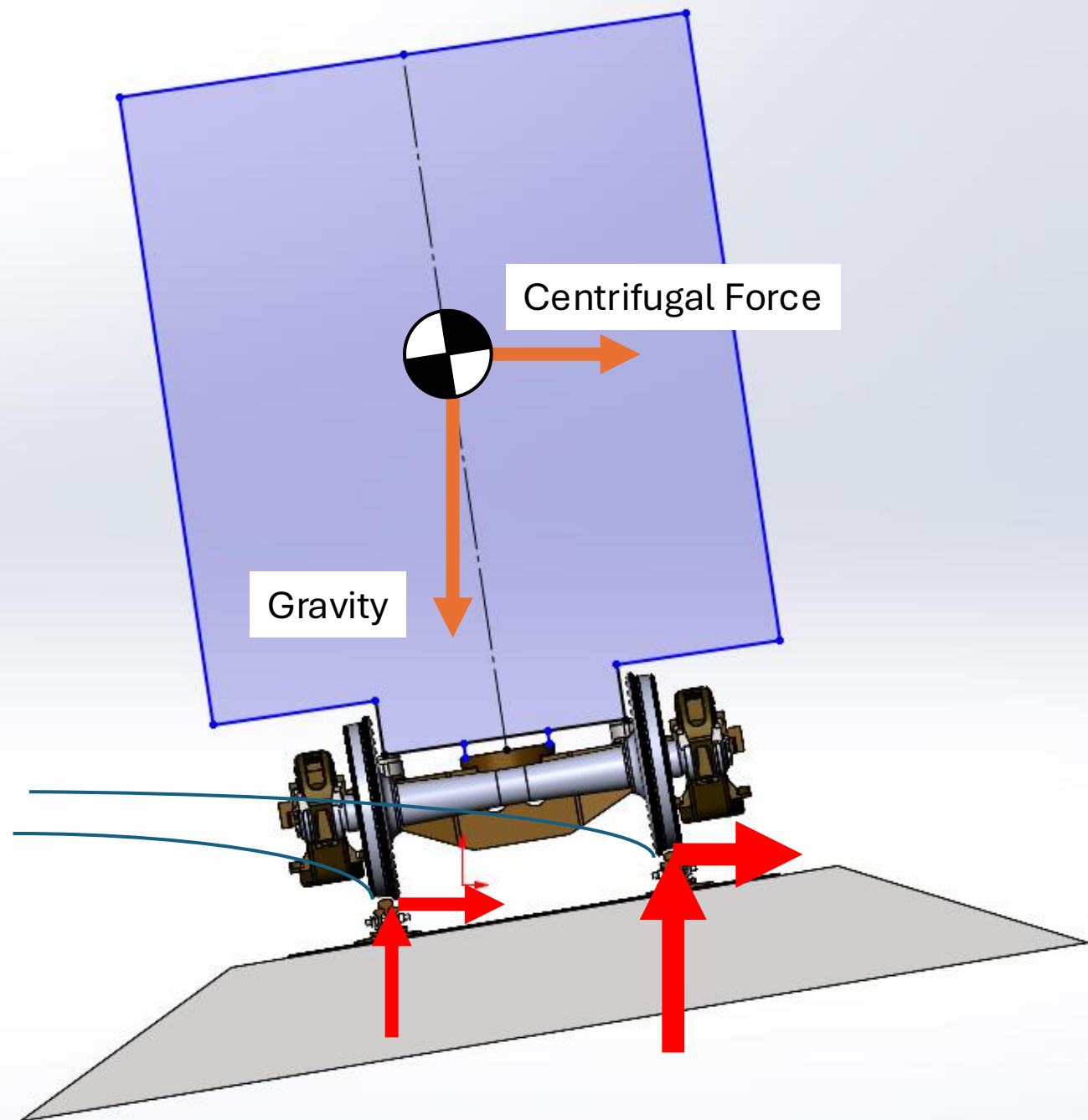


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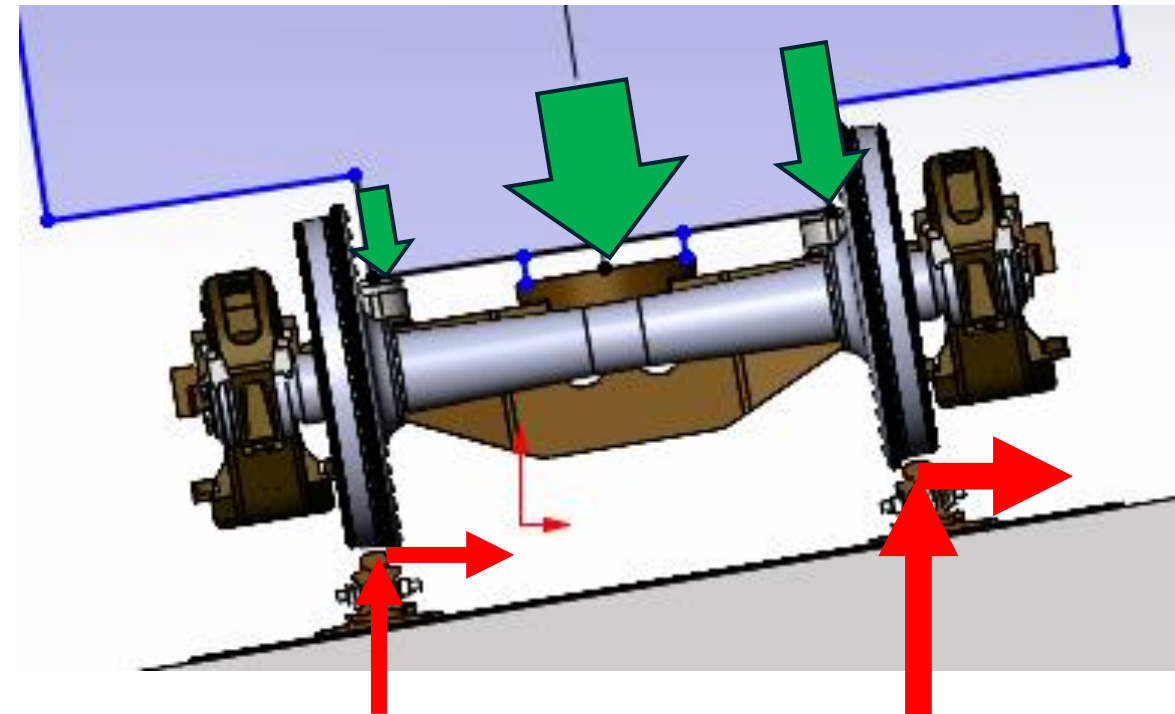


Steady-state curving

Curving left into the page at design speed



- **Expectation:** Navigate nominal irregularities and...
 - **Keep load upright**
 - Below the high rail at high speed
 - Above the rail at low speed
 - **Steer in the desired direction**
 - Keep truck square
 - Freely swivel





Summary

• Car Design

- Commodity
- Handling
- Clearance
- Infrastructure

• Suspension basics

- Wheels on rail
- Load upright
- Smooth and stable ride
- Wheels pointed in the right direction

• Trucks

- Simple design, complex details
- Suspension and steering
- Wheelsets do most of the steering
- Spring-mass-damper system

• Dynamics

- Resonance
- Empty / load behavior
- Damping
- Warp
- Tangent / Curving